

Space Charge Effects in Compact H⁻ Cyclotrons

1. Beam Blow-Up at Injection

- At injection (tens of keV), the beam is non-relativistic → strong Coulomb repulsion.
- The generalized perveance describes how strong space charge is:

where:

- I = beam current
 - γ = relativistic factors
 - At low injection energies, γ so becomes large → strong tune shifts, emittance growth.
 - **Result:** Beam bunches lengthen, transverse envelope oscillations increase, mismatch → particle loss.
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2. Longitudinal Space Charge

- Space charge forces lengthen bunches.
 - A longer bunch overlaps more RF phases → increases phase spread → particles drift out of acceleration bucket.
 - At high currents, neighboring bunches can even **interact** due to overlapping Coulomb fields.
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3. Transverse Space Charge & Tune Shift

- The focusing frequency (betatron tune) shifts downward with current:
 - If the tune approaches integer or half-integer resonances, strong beam blow-up and losses occur.
 - In compact cyclotrons, available focusing is limited, so this becomes a hard limit.
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4. Extraction Losses & Stripping

- For H^- cyclotrons, extraction is done via **stripper foils**.
- High current → higher foil heating → faster foil degradation.
- Space charge also enlarges the orbit spread at extraction → more particles miss the foil or hit wrong spots → increased losses.

Beam Dynamics Strategies to Mitigate Space Charge

Injection Stage

- **Higher injection energy:** Raising injection energy (e.g., from 30 keV to 70–100 keV) reduces perveance, hence weaker space charge.
- **Good matching:** Careful design of the central region (electrode shape, spiral inflectors, RF phase) to minimize mismatch.
- **Bunching:** Pre-bunching with RF before injection reduces longitudinal blow-up inside cyclotron.

Acceleration Stage

- **Strong RF focusing:** Higher harmonic cavities or higher dee voltage reduce bunch lengthening.
- **Large phase acceptance:** Optimized RF phase window reduces loss from longitudinal space charge blow-up.
- **Magnetic flutter / spiral sectors:** Improves transverse focusing strength.

Extraction Stage

- **Thin stripper foils:** Minimize heating, extend lifetime under high currents.
- **Orbit separation optimization:** Strong focusing in outer orbits helps separate turns, so space charge–induced spread is less critical.

Practical Beam Current Limits

- Commercial compact **medical isotope cyclotrons** (H^- type) typically deliver **0.5–1.5 mA** extracted beam.
- Attempts at **>2 mA** usually hit strong space charge blow-up and extraction losses.
- For comparison: **PSI's Injector-2** is not H^- , but with H_2^+ , they handle ~ 5 mA equivalent protons — showing what's possible if stripping/extraction challenges are bypassed.

🔑 So for compact H^- cyclotrons, the real **bottlenecks** are:

1. **Low-energy injection space charge** (limits capture efficiency).
2. **Longitudinal space charge blow-up** (limits phase acceptance).
3. **Foil heating and extraction losses** (limits extracted current).

Transverse: tune depression from space charge

1) Generalized perveance (how “strong” space charge is)

$$K = \frac{q I_{\text{peak}}}{2\pi\epsilon_0 m \gamma^3 \beta^3 c^3}$$

- $I_{\text{peak}} = I_{\text{avg}}/B_f$ with bunching factor $B_f \equiv I_{\text{avg}}/I_{\text{peak}}$ (typ. 0.25–0.5 in compact machines).
- **Scaling:** $K \propto I_{\text{peak}}/(\beta^3 \gamma^3)$. Raising injection energy ($\uparrow \beta \gamma$) or improving bunching ($\uparrow B_f \rightarrow \downarrow I_{\text{peak}}$) reduces K fast.

Quick feel: at 30 keV, $I_{\text{avg}} = 1$ mA, $B_f = 0.3 \Rightarrow I_{\text{peak}} \approx 3.3$ mA K is about $4\times$ larger than with the same current at 100 keV.

2) KV envelope & tune depression (smooth approximation)

For an rms beam radius R (round beam, KV-like),

$$R'' + \kappa(s) R - \frac{\epsilon^2}{R^3} - \frac{K}{R} \approx 0,$$

with external focusing κ and transverse (unnormalized) rms emittance ϵ .

Averaging over one cell (smooth focusing), the **depressed tune** satisfies roughly

$$\nu_{\perp} \approx \sqrt{\nu_0^2 - C_{\perp} \frac{K}{\varepsilon/R^2}},$$

where $C_{\perp}$ is an $\mathcal{O}(1)$ geometry factor that depends on sector shape/edge focusing and the actual envelope solution. The precise coefficient needs a lattice-specific calculation, but the **scaling** is robust:

- **Tune shift magnitude** $|\Delta\nu_{\perp}| \propto \frac{I_{\text{peak}}}{\beta^3\gamma^3} \frac{R^2}{\varepsilon}$ ($\times$ geometry).

Takeaways

- At fixed emittance, **doubling injection energy** (in keV) typically reduces $|\Delta\nu_{\perp}|$ by $\sim 30\text{--}50\%$ in the tens-of-keV regime (because β^3 rises quickly).
- At fixed energy, $|\Delta\nu_{\perp}|$ grows **linearly** with **peak current** and **quadratically** with beam size (since weaker external focusing $\rightarrow$ larger R , which hurts).
- At fixed I_{avg} , increasing B_f (more uniform current) **helps** by cutting I_{peak} .

3) Laslett-style form (aperture/image terms)

With a beam inside a round pipe of radius b and rms radius a , image forces amplify the shift. A common heuristic (from rings/coasting-beam theory, adapted for bunched beams) is

$$\Delta\nu_{\perp} \sim - \frac{I_{\text{peak}}}{\beta^3\gamma^3} \frac{G_{\perp}(b/a)}{\varepsilon_N} \left(\text{machine length scale} \right),$$

where $G_{\perp} \sim 1 + 2 \ln(b/a)$ for a round pipe and $\varepsilon_N = \beta\gamma \varepsilon$ is the normalized emittance. The exact length-scale coefficient depends on your focusing lattice and where in the machine you evaluate it. The **scaling** messages remain:

- Worse when **beam is large** vs **pipe** (small $b/a \rightarrow$ bigger $G_{\perp}$).
- Improves with **normalized emittance** and with $\beta^3\gamma^3$.

Rule of thumb for limits: try to keep $|\Delta\nu_{\perp}| \lesssim 0.2\text{--}0.3$ to avoid resonance overlap in compact cyclotrons. That target, plugged into the scalings above, gives a quick I_{max} estimate.

Longitudinal: bunch lengthening from space charge

In an isochronous cyclotron, “synchrotron motion” is weak; **RF phase focusing** and the finite **RF gap transit-time factor** provide the restoring force. Space charge produces an **inductive** longitudinal impedance that **lengthens** the bunch (potential-well distortion).

1) Effective space-charge impedance (low frequency)

For a round beam in a conducting pipe, the effective **longitudinal** space-charge impedance per harmonic n scales like

$$\left| \frac{Z_{\parallel}}{n} \right| \propto \frac{Z_0}{2\pi} \frac{G_{\parallel}(b/a)}{\beta \gamma^2}, \quad Z_0 = 377 \, \Omega,$$

with $G_{\parallel}(b/a) \sim 1 + 2 \ln(b/a)$. **Scaling:** $\propto I_{\text{peak}}/(\beta \gamma^2)$, i.e., softer than transverse's $1/(\beta^3 \gamma^3)$ but still very sensitive at low energy.

2) Balance with RF phase focusing

Let the **RF focusing constant** (small-amplitude) scale as

$$k_{\phi} \propto \frac{h q V_{\text{rf}} |\cos \phi_s|}{E_{\text{kin}}},$$

with harmonic h , dee voltage V_{rf} , synchronous phase ϕ_s , and kinetic energy E_{kin} .

In a **potential-well** picture, the **equilibrium rms bunch length** (radians) approximately follows a **1/3-power law**:

$$\sigma_{\phi} \propto \left[\frac{I_{\text{peak}}}{\beta \gamma^2} G_{\parallel}(b/a) \bigg/ \frac{h q V_{\text{rf}} |\cos \phi_s|}{E_{\text{kin}}} \right]^{1/3}.$$

Scaling highlights

- $\sigma_{\phi} \uparrow$ with I_{peak} and with $G_{\parallel}$ (tighter aperture makes it worse).
 - $\sigma_{\phi} \downarrow$ with V_{rf} , $|\cos \phi_s|$, h , and E_{kin} .
 - Because of the 1/3 power, **doubling** I_{peak} increases the **bunch length** by only $\approx 26\%$, but that's enough to spill particles outside the stable RF phase window and seed losses.
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Coupling: why longer bunches hurt transverse limits

Longer bunches $\rightarrow$ smaller bunching factor $B_f \rightarrow$ larger $I_{\text{peak}} \rightarrow$ bigger $K \rightarrow$ more transverse tune depression. This positive feedback is why compact machines often hit a sharp current wall: once bunches lengthen past the RF bucket edges, both transverse and longitudinal losses snowball.

Practical “dials” and their math

- **Raise injection energy** E_{inj} : $|\Delta\nu_{\perp}| \sim I_{\text{peak}}/(\beta^3\gamma^3)$ and $\sigma_{\phi} \sim (I_{\text{peak}}/(\beta\gamma^2))^{1/3}$. Even modest E_{inj} increases (30 $\rightarrow$ 70 keV) can **halve** transverse tune shift and **shrink** longitudinal impedance meaningfully.
 - **Improve B_f** (pre-buncher, better central-region capture): $\Downarrow I_{\text{peak}} \rightarrow$ linear relief to both transverse and longitudinal space charge.
 - **Increase V_{rf}** or operate nearer $\phi_s \simeq 0$ (as allowed by capture/extraction): $\sigma_{\phi} \propto (1/V_{\text{rf}})^{1/3}$. Also higher V_{rf} raises turn-to-turn energy gain, improving turn separation at extraction.
 - **Keep beam small vs aperture** (large b/a , better matching): Both $G_{\perp}$ and $G_{\parallel}$ decrease with larger b/a .
 - **Preserve normalized emittance ε_N** (good matching through the spiral inflector, avoid mismatch oscillations): Larger ε_N tolerates more current before $|\Delta\nu_{\perp}|$ reaches danger.
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Worked “feel” numbers (just to anchor intuition)

Consider H^- , $I_{\text{avg}} = 1 \text{ mA}$, $B_f = 0.3 \Rightarrow I_{\text{peak}} \approx 3.3 \text{ mA}$.

- **Perveance K** :
 - 30 keV $\rightarrow K \approx 4.1 \times 10^{-4}$

- 70 keV $\rightarrow K \approx 1.2 \times 10^{-4}$ (i.e., $\sim 3.5\times$ lower by 70 keV at the same average current and B_f)

- **Bunch length trend:** With unchanged RF, $\sigma_\phi \propto (I_{\text{peak}}/(\beta\gamma^2))^{1/3}$, so going 30 $\rightarrow$ 70 keV cuts the **space-charge drive** by roughly a factor $\sim (\beta\gamma^2)^{-1/3} \rightarrow$ expect **$\sim 20\text{--}30\%$ shorter** equilibrium bunch length solely from the energy term, before any RF changes.

(Coefficients depend on your exact central region, aperture, harmonic, and RF focusing; use these as *scalings*, not hard predictions.)

How to use these scalings to set an I_{max}

1. Pick an acceptable tune depression, e.g. $|\Delta\nu_\perp| \lesssim 0.25$.
 2. Use the transverse scaling $|\Delta\nu_\perp| \propto (I_{\text{peak}}/(\beta^3\gamma^3)) (R^2/\varepsilon) G_\perp$ to solve for $I_{\text{peak,max}}$.
 3. Check **longitudinal**: with that current, compute σ_ϕ from the $1/3$ law; ensure it still fits in your **stable RF phase window** and doesn't degrade B_f so much that step 2 breaks.
 4. Iterate with E_{inj} , V_{rf} , B_f , **matching**.
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(what grows fastest and what helps most)

- Bad guys: I_{peak} , small E_{inj} (low $\beta\gamma$), tight aperture b , large beam a , low V_{rf} , poor capture (small B_f).
- Good guys: raise E_{inj} (huge win; transverse scales as $1/(\beta^3\gamma^3)$), improve B_f , increase V_{rf} , keep b/a big, preserve ε_N .

Transmission Efficiency

For typical compact H^- cyclotrons with axial injection, **the stand-alone spiral-inflector transmission is often only a few–tens of percent**; a realistic back-of-the-envelope gives $\sim 5\text{--}15\%$, and after including central-region capture it can drop to $\sim 1\text{--}5\%$ **overall** from source current to first accelerated turns.

Quick estimate (order-of-magnitude)

Take representative numbers for a 30 keV H^- beam:

- rms **normalized emittance** from the source & LEBT: $\varepsilon_{N,\text{rms}} \sim 5 \pi \text{ mm mrad}$ (often 5–15)
- $\beta\gamma \simeq 0.008$ at 30 keV $\rightarrow$ **geometric emittance** $\varepsilon_{\text{rms}} = \varepsilon_N / (\beta\gamma) \approx 5 / 0.008 \approx 625 \pi \text{ mm mrad}$.
- A compact spiral inflector with $\sim 5 \text{ mm}$ gap, length $\approx 3\text{--}5 \text{ cm}$, $|V| \approx 10\text{--}20 \text{ kV}$ in $B \sim 1.1\text{--}1.4 \text{ T}$ has a **geometric acceptance** typically of order $A_{\text{inf}} \sim 30\text{--}80 \pi \text{ mm mrad}$ (machine-dependent).

Transmission (emittance-limited) $\approx A_{\text{inf}} / \varepsilon_{\text{rms}}$:

$$T_{\text{inflector}} \sim \frac{30\text{--}80}{625} \approx 5\%\text{--}13\%.$$

Even if your source/LEBT is excellent (e.g., $\varepsilon_{N,\text{rms}} = 3 \pi$), you still get only $\sim 10\%\text{--}20\%$.

Once you include **RF phase acceptance** of the first turns (often only $20\text{--}40^\circ$ out of 360°) and central-region losses, **source-to-capture** efficiency can fall to $\sim 1\text{--}5\%$.

Why the transmission is intrinsically small

1. Emittance vs. small aperture (hard acceptance mismatch)

- The **spiral-inflector gap** is tiny ($3\text{--}6 \text{ mm}$) and its length is short.
- That sets a **small transverse acceptance** compared with the **large geometric emittance** you get when you divide a modest normalized emittance by tiny $\beta\gamma$ at 30 keV.
- Bottom line: most particles simply don't fit the accepted phase-space volume.

2. Space-charge blow-up in the inflector (no neutralization)

- In the LEBT you often have partial neutralization; **inside the inflector** the beam is between electrodes $\rightarrow$ **neutralization is lost**.
- At 30 keV, $\beta, \gamma \ll 1 \rightarrow$ generalized perveance $K \propto I / (\beta^3 \gamma^3)$ is large; the bunch **expands** and scrapes the plates.

3. Strong, nonlinear fields and tight tolerances

- The superposed **E (inflector)** and **B (cyclotron)** fields generate curved 3D trajectories; fringe fields at entrance/exit add **nonlinear focusing**.
- **Plate edges** and finite gap produce field errors; $100 \mu\text{m}$ misalignments can cause mm-level orbit shifts $\rightarrow$ scraping.

4. Mismatch into the median-plane optics

- The beam exiting the inflector is usually **astigmatic** with significant **x-z/y-z coupling**.
- The very first gaps offer little focusing/phase control → envelope oscillations → losses on posts, dees, or puller.

5. Energy and phase spread from the inflector

- Path-length and fringe-field variations convert into **energy/phase spread**, reducing the fraction that lands inside the **RF phase bucket**.

6. H^- -specific issues

- **Lorentz stripping margins** limit how high you can push B and plate voltage (E-field), which in turn caps the compactness/acceptance of the device.
- Any grazing contact with metal or residual gas collision can **strip** H^- → immediate loss.

What helps (in order of impact)

- **Raise injection energy** (e.g., 30 → 60–80 keV): geometric emittance shrinks as $1/(\beta\gamma)$ and space-charge weakens as $1/(\beta^3\gamma^3)$.
- **Reduce source emittance** (better extraction optics, solenoid matching, apertures).
- **Pre-bunch & match** to the inflector using LEBT optics; keep bunching factor high to lower peak current.
- **Larger gap / optimized 3D inflector** (field-mapped design, shaped electrodes) to increase acceptance without exceeding E-field limits.
- **Precise alignment** (≤ 50 – $100\ \mu\text{m}$) and smooth edges to cut nonlinearities.
- **Central-region redesign** for stronger early focusing and larger RF phase acceptance.

Estimate for a **compact proton (H^+) cyclotron** with **axial injection** and a **neutralizing plasma column** just upstream of the spiral inflector.

- **Energy** $E = 30\ \text{keV} \rightarrow \beta \simeq 0.007996, \gamma \simeq 1.000032$.
- **Normalized rms emittance** $\varepsilon_{N,\text{rms}} = 5\ \pi\ \text{mm-mrad} \Rightarrow$ geometric rms emittance $\varepsilon_{\text{rms}} = \varepsilon_N/(\beta\gamma) \approx 625\ \pi\ \text{mm-mrad}$.
- **Column length** $L = 0.2$ – $0.3\ \text{m}$, **pressure** $\approx 10^{-5}\ \text{torr}$.
- **Current** $I = 1$ – $10\ \text{mA}$.
- Use a **typical spiral-inflector acceptance** $A_{\text{inf}} \sim 60\ \pi\ \text{mm-mrad}$ and geometry (few-mm gap, $|V| \sim 10$ – $20\ \text{kV}$).

1) What neutralization changes (and doesn't)

Neutralization fraction η reduces the **effective perveance**:

$$K_{\text{eff}} = (1 - \eta) K.$$

In the space-charge dominated regime, the **matched envelope** (size and divergence) scales roughly as

$$R, R' \propto \sqrt{1 - \eta}.$$

The **normalized emittance itself is not improved**; you are removing space-charge defocus so the **projected beam** entering the inflector is smaller and better matched.

A very good rule of thumb for **inflector transmission gain** is

$$\text{gain} \approx \frac{1}{\sqrt{1 - \eta}}$$

(until you hit the inflector's hard aperture and central-region limits).

2) How much neutralization at your gas/length?

At 10^{-5} torr and $L = 0.2 - 0.3$ m, the beam's own ionization readily supplies electrons; build-up times are tens of μs (faster for Kr). Reasonable steady η ranges:

- Hydrogen (H_2): $\eta \approx 0.70 - 0.85$
- Krypton (Kr): $\eta \approx 0.85 - 0.95$

(Heavier Kr has larger ionization cross-section and holds electrons more effectively, pushing η higher. At these pressures and path lengths, multiple scattering is negligible.)

Current dependence: at higher beam current the **residual** space charge $(1 - \eta)I$ is larger; η can sag slightly with I if electrons are over-cleared. Below, I assume a mild trend with current.

3) Envelope “focusing” before the inflector

Relative *shrink* of beam size/divergence vs. un-neutralized:

$$\frac{R_\eta}{R_0} = \sqrt{1 - \eta}.$$

Gas	η (low–high)	Size/divergence shrink R_η/R_0
H ₂	0.70 → 0.85	0.548 → 0.387
Kr	0.85 → 0.95	0.387 → 0.224

So with Kr you can expect the projected envelope at the inflector entrance to be ~2.6–4.5× smaller area (since area $\propto R^2$) than without neutralization.

4) Transmission improvement through the inflector

Baseline, **un-neutralized** acceptance estimate:

$$T_0 \approx \frac{A_{\text{inf}}}{\epsilon_{\text{rms}}} \approx \frac{60}{625} \approx 9.6\% \quad (\text{order of magnitude}).$$

Improved transmission $T \approx T_0 \times \text{gain}$, with $\text{gain} = 1/\sqrt{1 - \eta}$. I also show a gentle $\eta(I)$ trend:

H₂ column (30 keV)

Beam current	assumed η	gain	$T \approx T_0 \times \text{gain}$
1 mA	0.80	2.236	21%
5 mA	0.75	2.000	19%
10 mA	0.70	1.826	18%

Kr column (30 keV)

Beam current	assumed η	gain	$T \approx T_0 \times \text{gain}$
1 mA	0.92	3.536	34%
5 mA	0.90	3.162	30%
10 mA	0.85	2.582	25%

These are **stand-alone inflector transmission** estimates; **overall source→capture** will be lower once RF phase acceptance and central-region losses are included. Also, if the inflector's **hard aperture** becomes the limiter, transmissions will **saturate** around ~30–50% even if η is increased further.

5) Sanity checks (your conditions)

- **Neutralization time:** with 10^{-5} torr and 30 keV protons, build-up times $\sim 25\text{--}130\ \mu\text{s}$ (Kr faster than H_2) — easily fast enough for CW beams.
- **Scattering/emittance growth:** negligible at 10^{-5} torr over 0.2–0.3 m.
- **Downstream matching:** once neutralized, **re-match LEBT optics** so the smaller envelope actually lands in the inflector gap; otherwise you leave performance on the table.

Bottom line

- Expect **envelope shrink** by a factor $0.55 \rightarrow 0.39$ (H_2) and $0.39 \rightarrow 0.22$ (Kr) relative to un-neutralized, at your pressure/length.
- That translates to about $2\times$ (H_2) up to $3\text{--}4.5\times$ (Kr) **gain in inflector transmission**, i.e., ~18–21% (H_2) and ~25–34% (Kr) for your 30 keV, 1–10 mA range — assuming a typical inflector acceptance and good matching.